

IMU Noise and Performance Characteristics

Introduction

The Inertial Measurement Unit (IMU) represents a cornerstone in modern navigation, motion tracking, and orientation detection technologies. Among these devices, the VN 100, developed by VectorNav Technologies, stands out as a prominent example. This sophisticated sensor module combines gyroscopes, accelerometers, and magnetometers to provide accurate and reliable measurements of angular velocity, linear acceleration, and magnetic field vectors. Designed for a wide range of applications, from unmanned vehicle navigation to sports science, the VN 100 stands out for its precision and robustness. Key to its performance is the sensor fusion algorithm, which integrates data from multiple sensors to provide a comprehensive view of an object's orientation and movement. This feature is crucial in ensuring data accuracy and reliability, especially in dynamic environments. This report focuses on analyzing data from the VN 100 under various conditions to evaluate its noise characteristics and performance. The insights gained will shed light on the VN 100's capabilities and its applicability in real-world scenarios.

Analysis

a. 5-Hour Stationary Data Analysis

The bag file assigned to me is **Location A**. This data set likely reflects the baseline noise characteristics of the VN 100 in a relatively controlled environment. The noise parameters for the gyroscope and accelerometer indicate the inherent stability and precision of the sensor under stationary conditions. The gyroscope noise parameters show low values for N, K, and B, indicating good stability over the 5-hour period. The accelerometer noise parameters also exhibit relatively low noise levels, suggesting minimal interference from environmental factors. The low noise levels in this data set are indicative of the IMU's inherent electronic noise and sensor stability under stationary conditions.

Noise Parameters	Angle Random Walk (N)	Rate Random Walk (K)	Bias Instability (B)
Gyroscope X	9.0084e-5	3.9555e-7	7.1553e-6
Gyroscope Y	1.1251e-4	3.5289e-7	1.0239e-5
Gyroscope Z	9.1164e-5	4.7209e-5	9.446e-5
Accelerometer X	1.9665e-3	2.7509e-5	8.5466e-4
Accelerometer Y	2.1437e-3	4.4462e-5	5.0139e-4
Accelerometer Z	2.7405e-3	1.2282e-4	7.9223e-4

The comparison of the data values with the VN 100 data sheet shows that the values are comparable to that of a Navigation grade sensor.

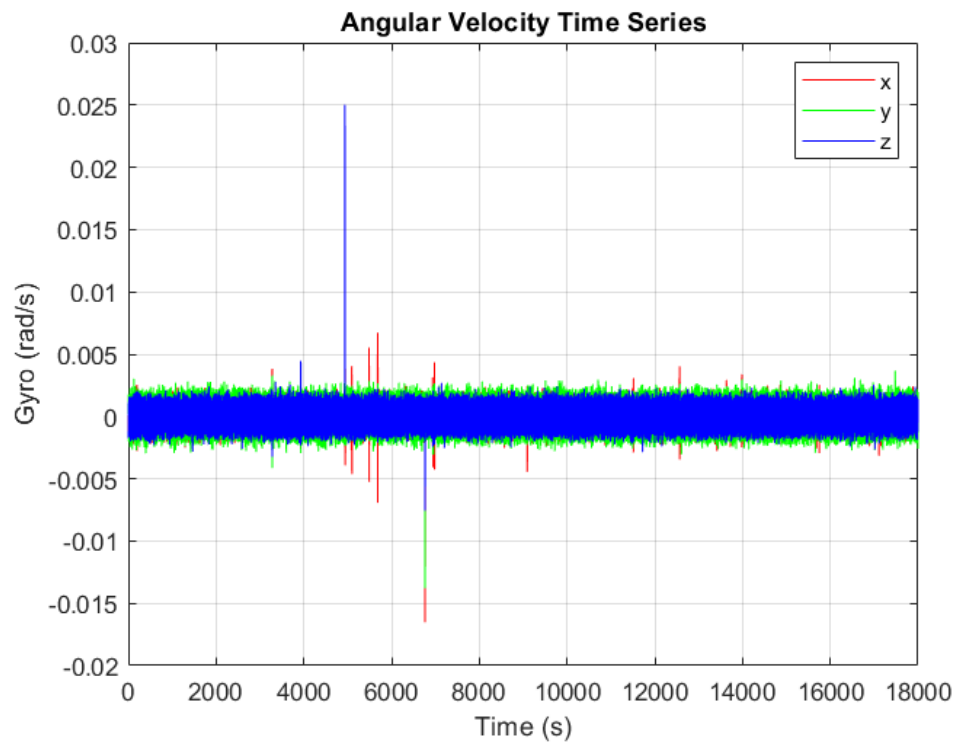
Error Terms by Sensor Grade

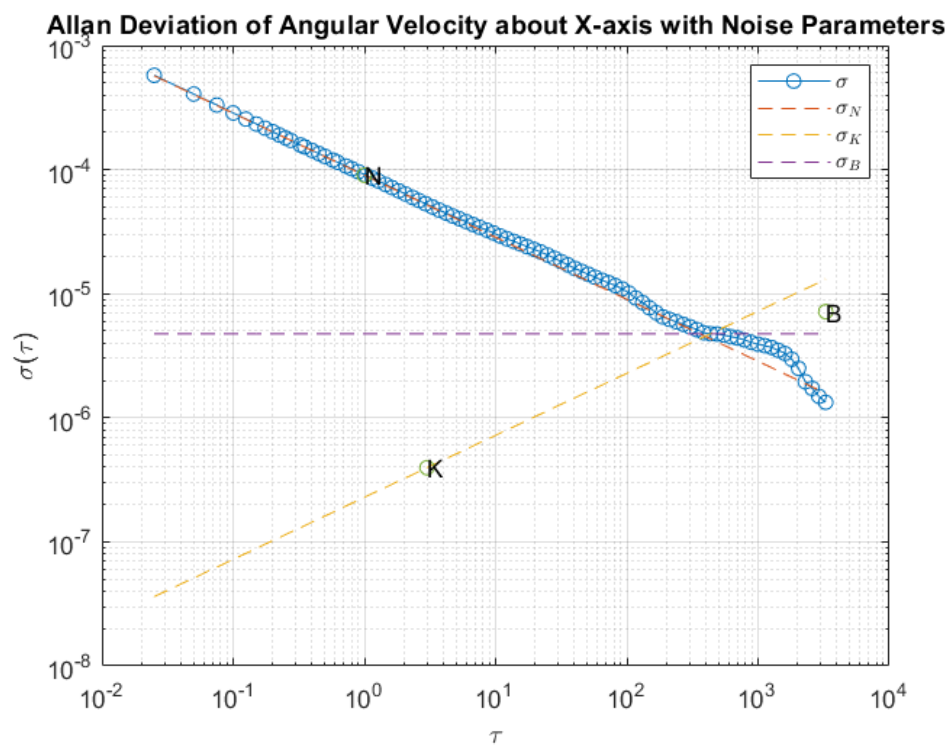
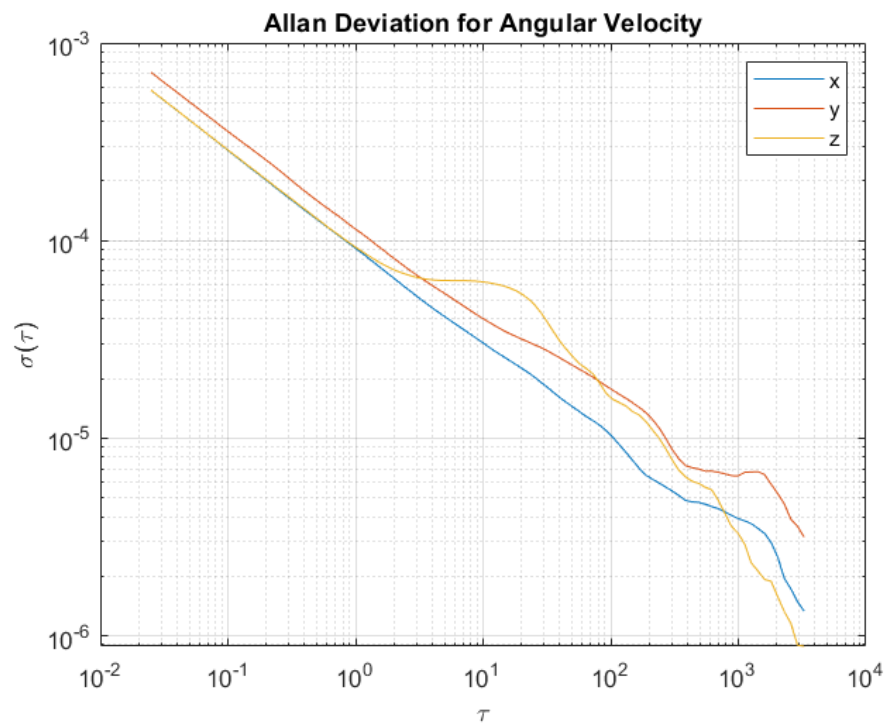
GRADE	ACCELEROMETER BIAS (mg)	VELOCITY RANDOM WALK (m/s/ $\sqrt{\text{hr}}$)	GYRO BIAS (deg/hr)	ANGLE RANDOM WALK (deg/ $\sqrt{\text{hr}}$)
Consumer	10	1	100	2
Industrial	1	0.1	10	0.2
Tactical	0.1	0.03	1	0.05
Navigation	0.01	0.01	0.01	0.01

The potential source of error includes:

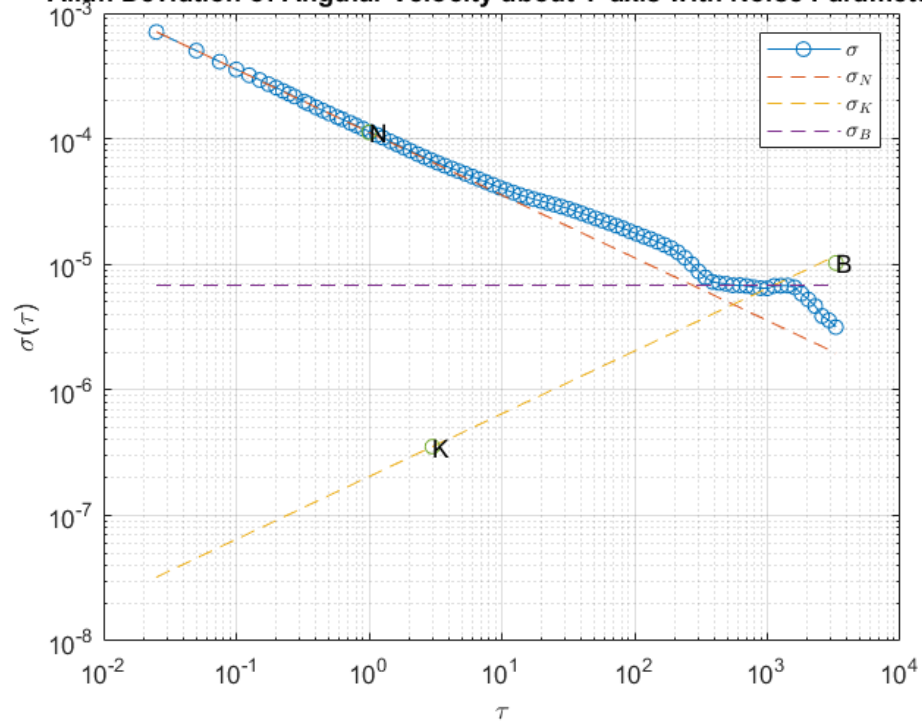
- Environmental factors - Long duration stationary measurements are susceptible to environmental noise such as temperature fluctuations, which can affect sensor readings.
- Electronic noise - Inherent electronic noise in the sensor circuitry may become more pronounced in long-duration stationary data.
- Drift - Over extended periods, sensors like gyroscopes and accelerometers may exhibit drift, which is a slow change in the sensor output when there is no actual change in the measured quantity.

Based on the characteristics of the AVAR plots, the noise parameters and comparisons with my teammates plots and parameters, I presume that the data in Location A bag was collected from the fifth floor of the ISEC building.

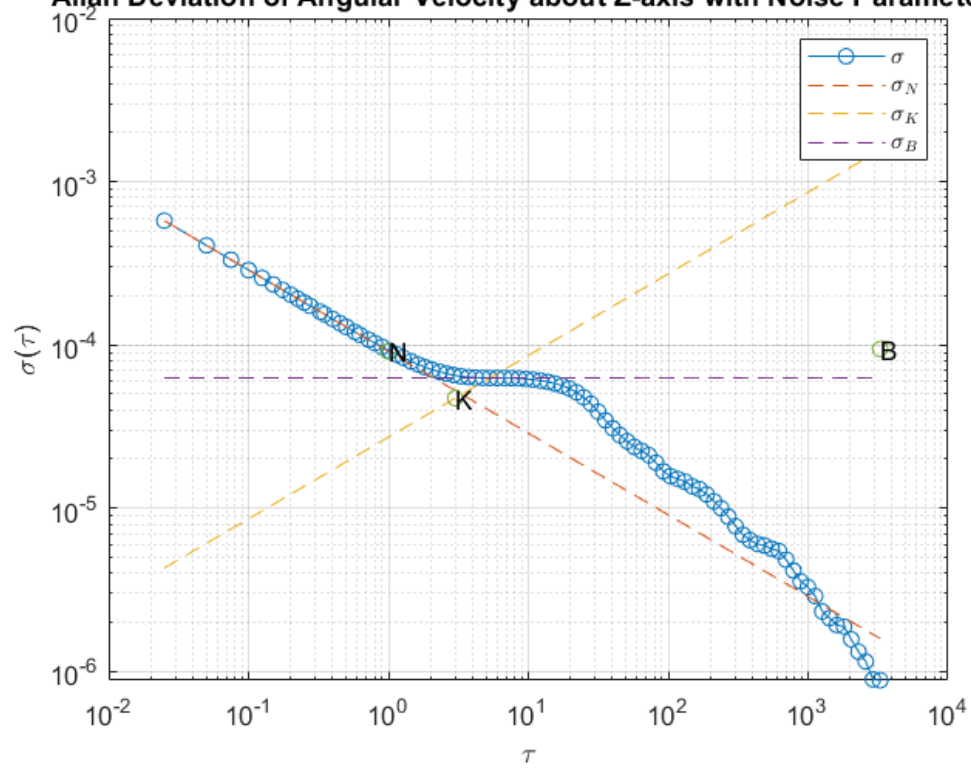


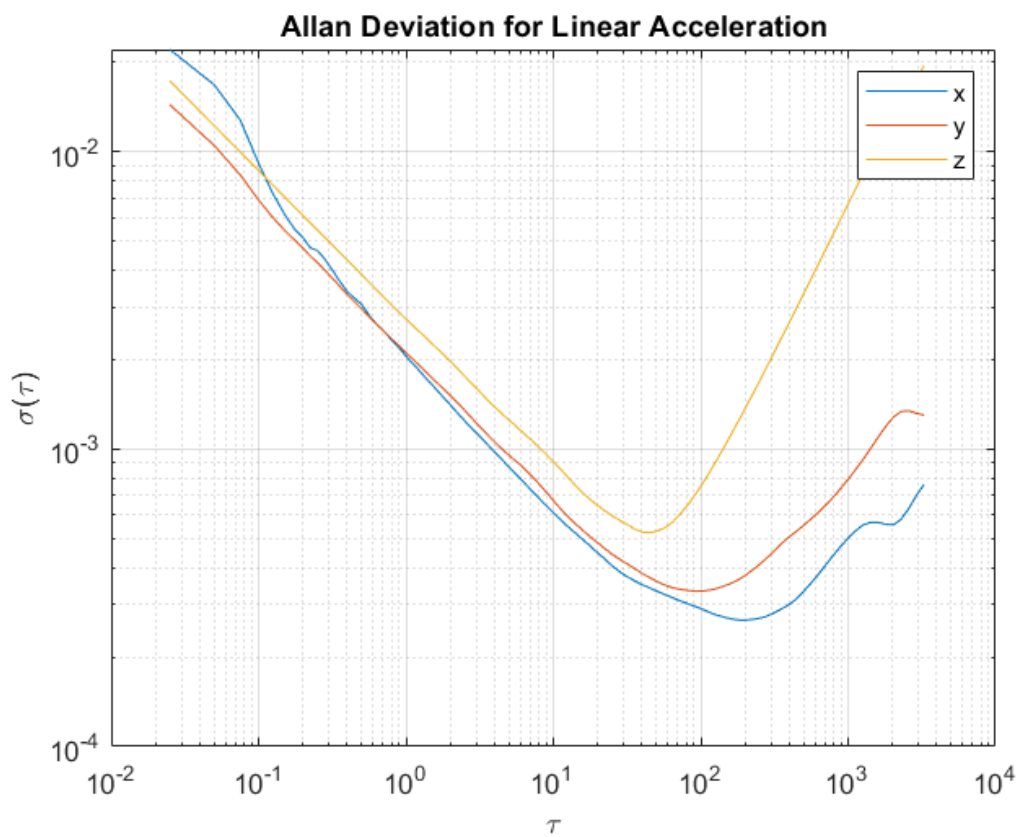
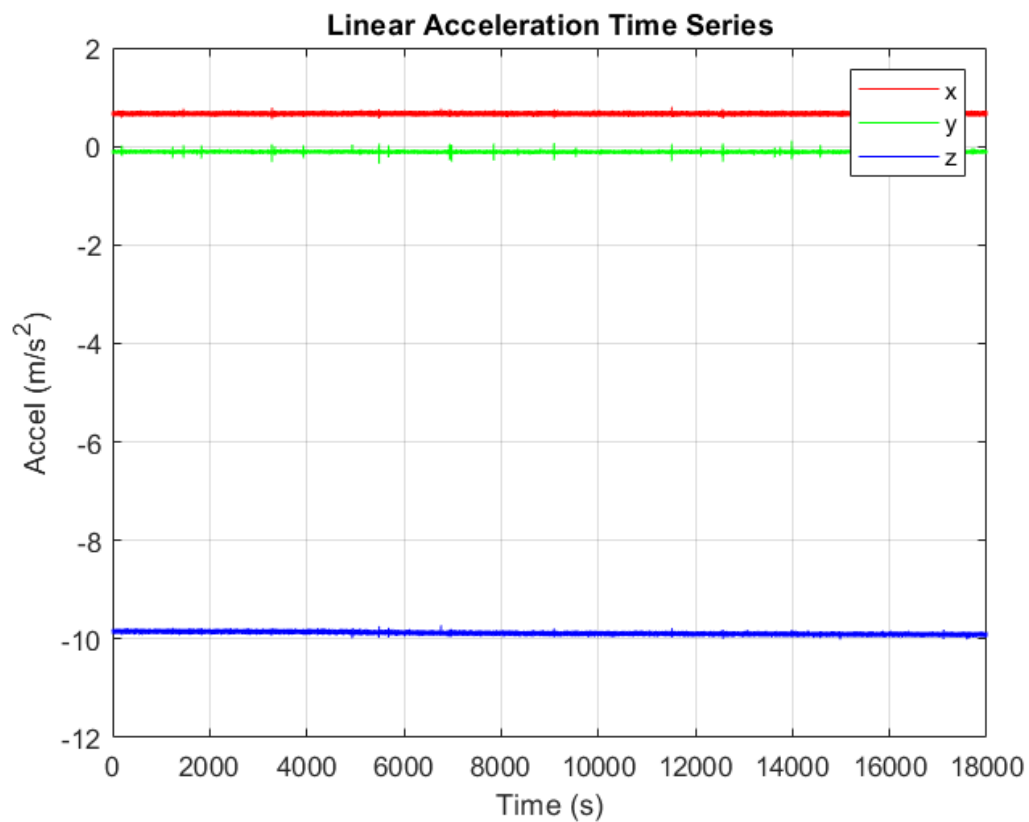


Allan Deviation of Angular Velocity about Y-axis with Noise Parameters

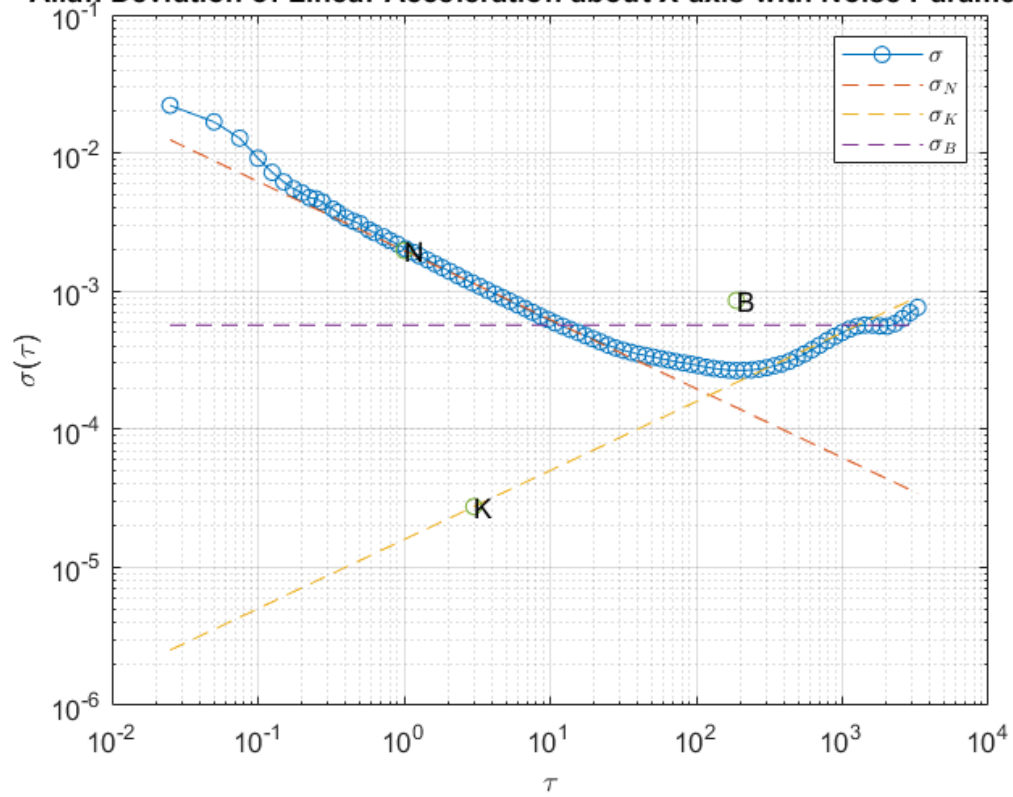


Allan Deviation of Angular Velocity about Z-axis with Noise Parameters

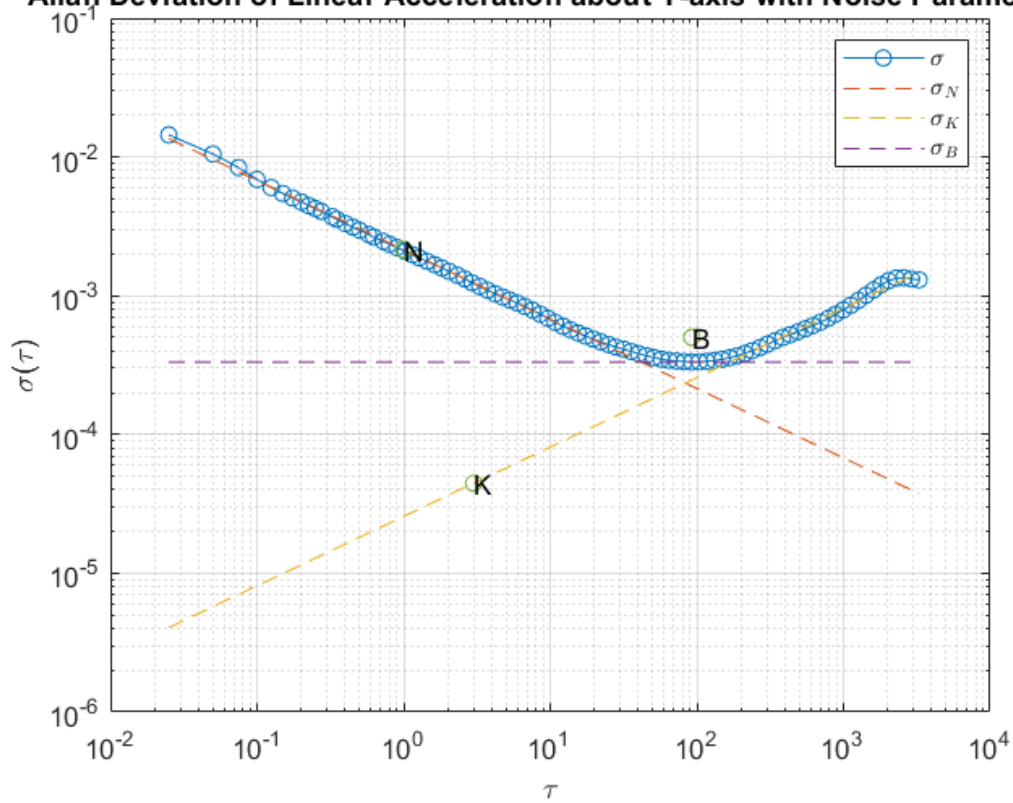


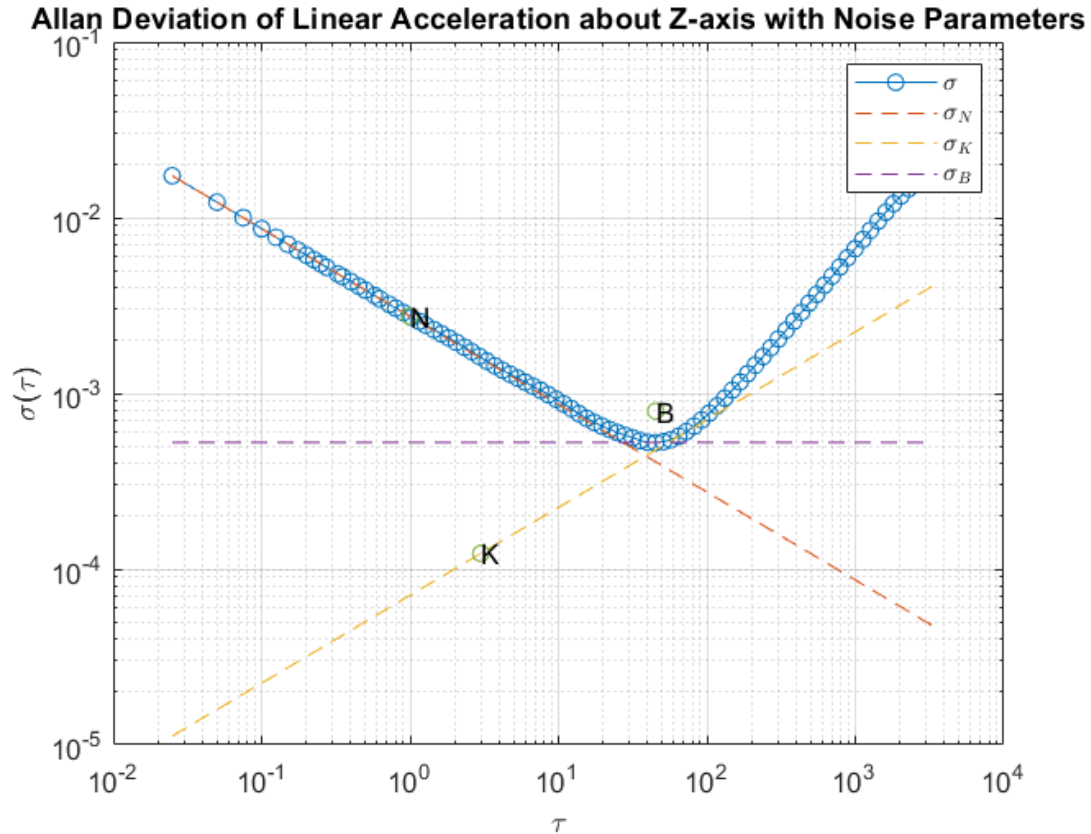


Allan Deviation of Linear Acceleration about X-axis with Noise Parameters



Allan Deviation of Linear Acceleration about Y-axis with Noise Parameters



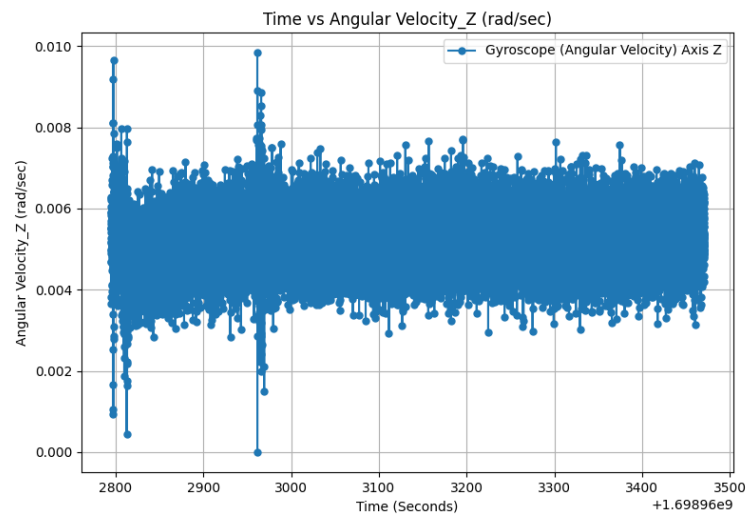
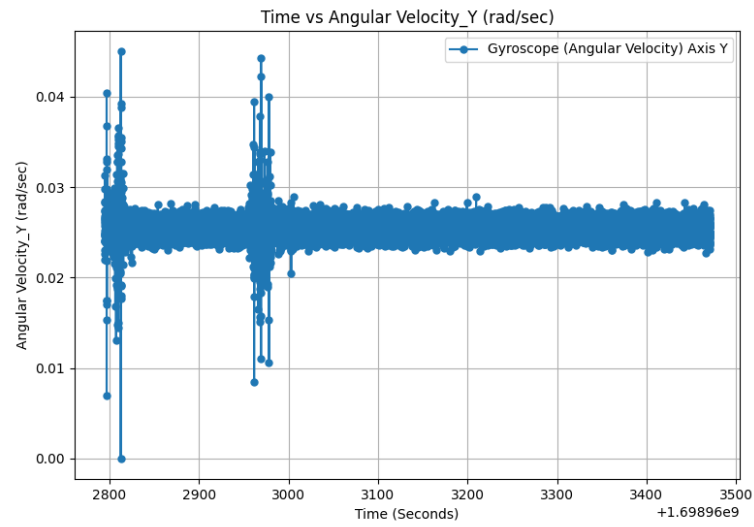
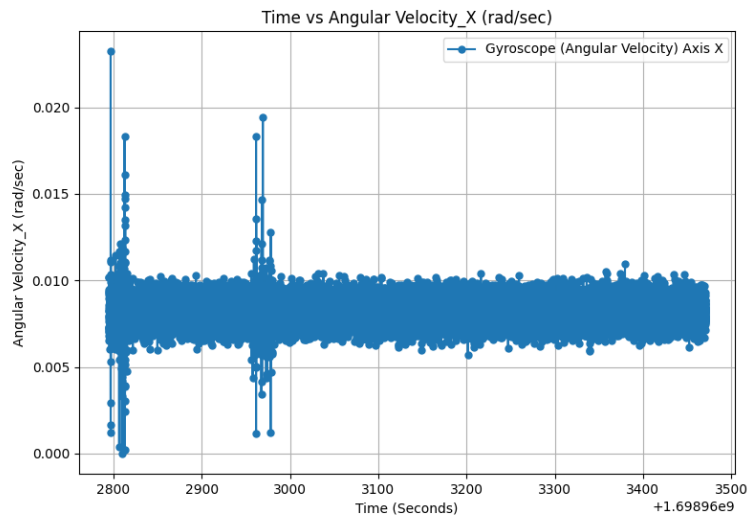


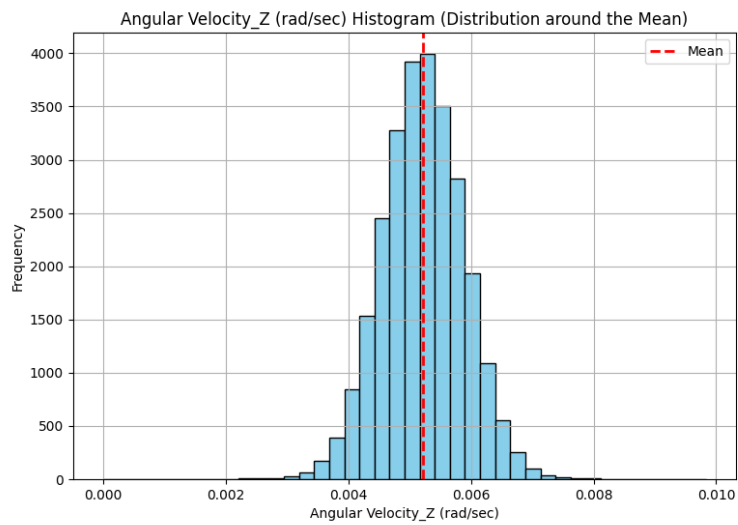
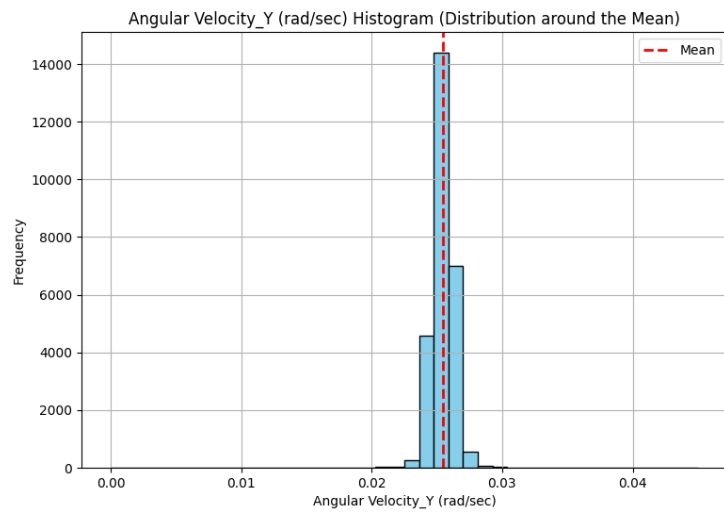
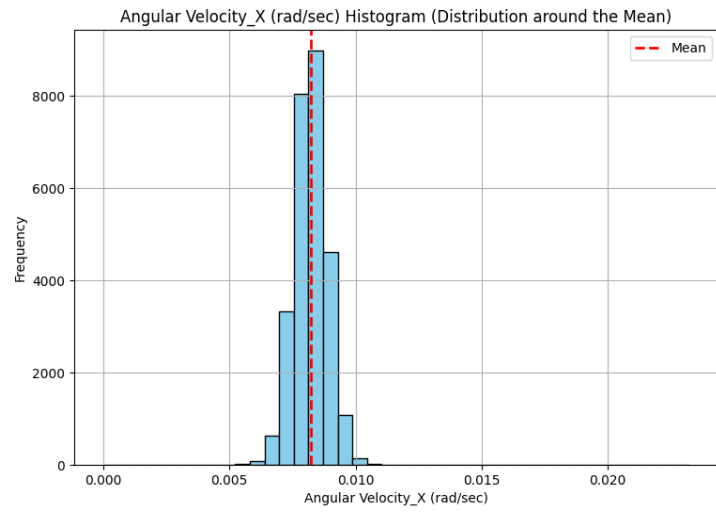
b. 5-minute Stationary Data Analysis

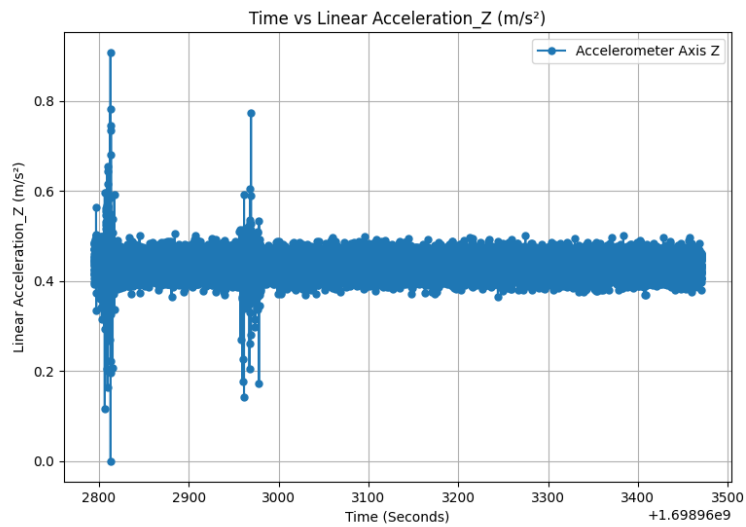
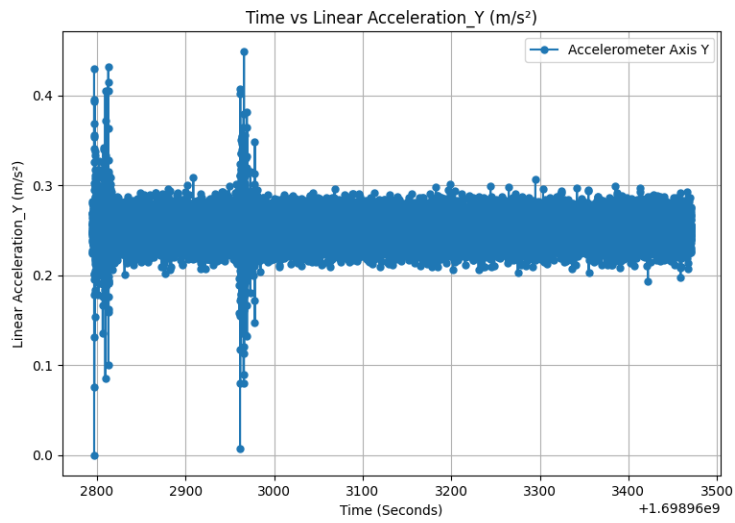
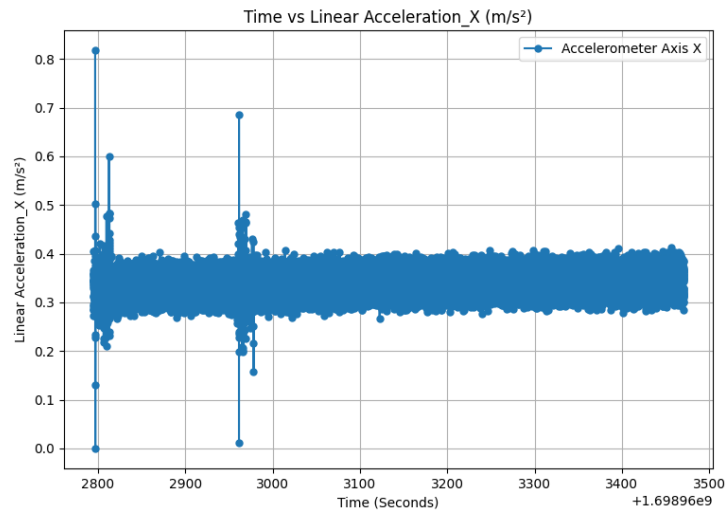
The dataset was collected near the Khoury buildings. This shorter stationary data set provides insight into the sensor's performance over a brief period, where environmental and operational conditions can have a more pronounced impact. The gyroscope and accelerometer statistics are consistent with a stable, stationary environment but may capture minor vibrations or movements not present in the 5-hour data. The data mostly conforms to expected Gaussian noise characteristics, with some deviations possibly due to external factors like minor movements or electronic fluctuations. The low mean and median values indicate relatively stable and predictable noise characteristics. The magnetometer statistics suggest minimal external magnetic interference in the short stationary setting. This dataset likely captures both the inherent sensor noise and minor environmental fluctuations.

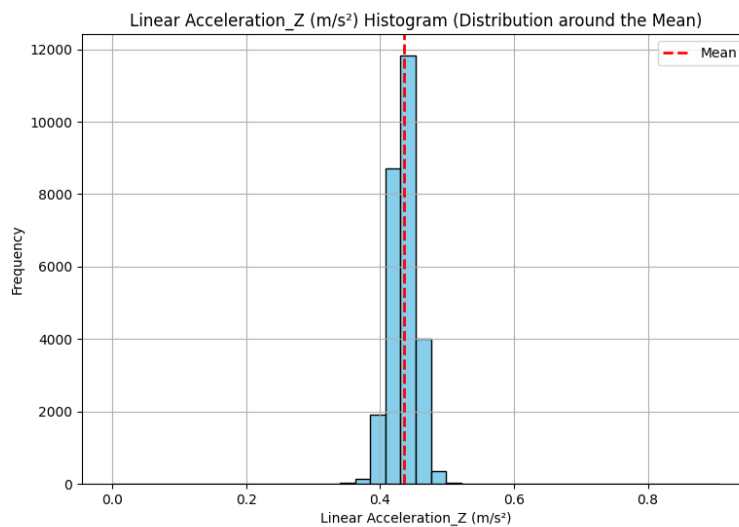
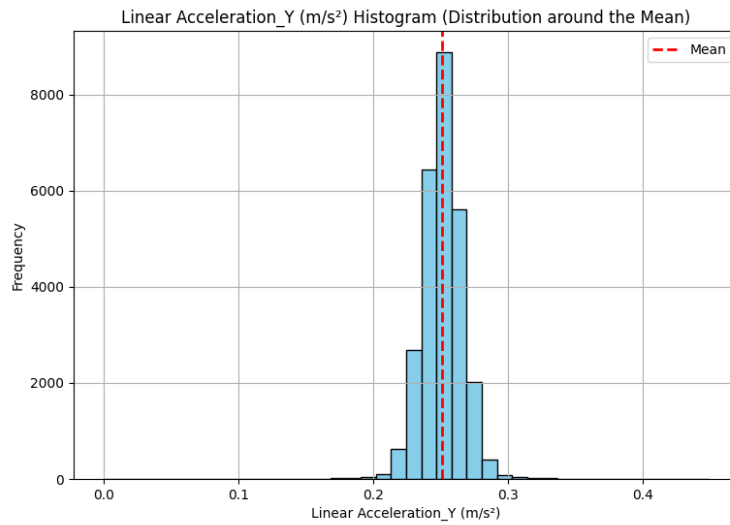
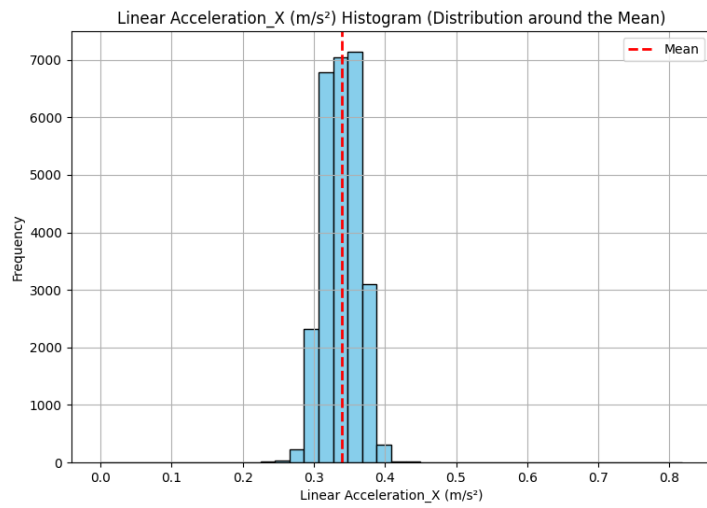
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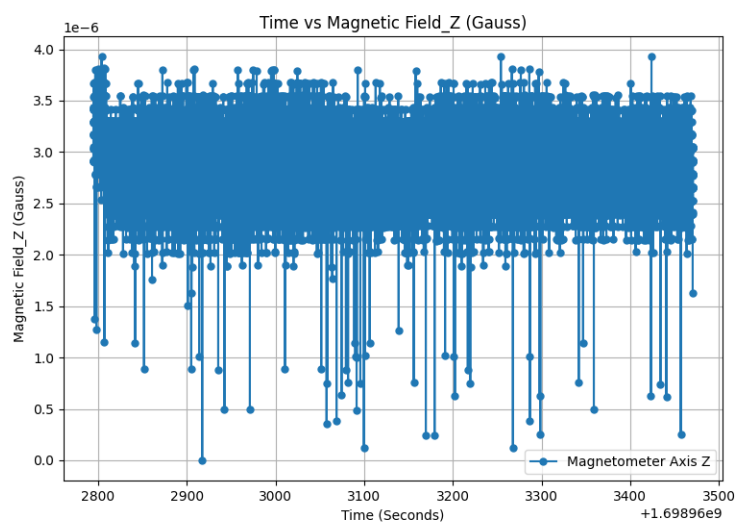
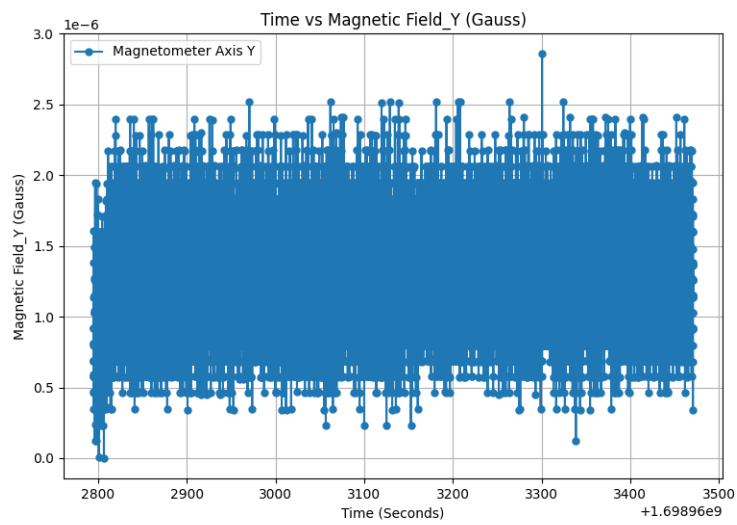
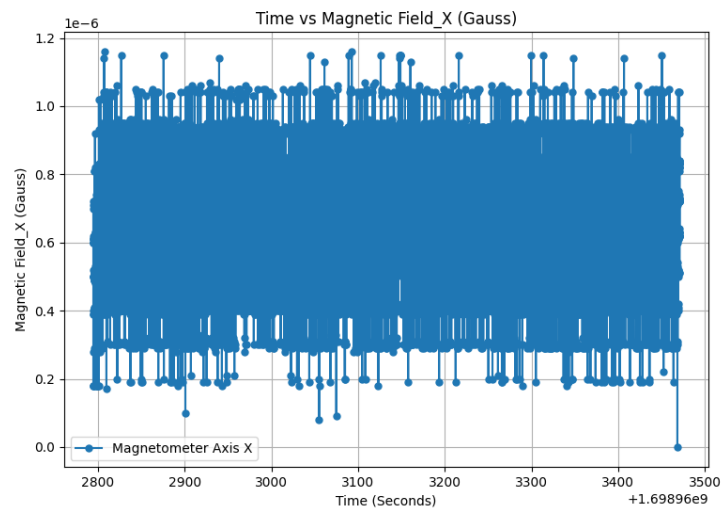
- **Minor Movements** - Even when stationary, minor vibrations, or movements (e.g., from nearby foot traffic, vehicles, tram and building vibrations) can introduce noise.
- **Calibration Errors** - Shorter data collection periods might not fully account for calibration errors, which can introduce bias in the readings.
- **Sensor Sensitivity** - The IMU's sensitivity to minor fluctuations might be more noticeable in short-duration data, especially if the sensor is high-precision.

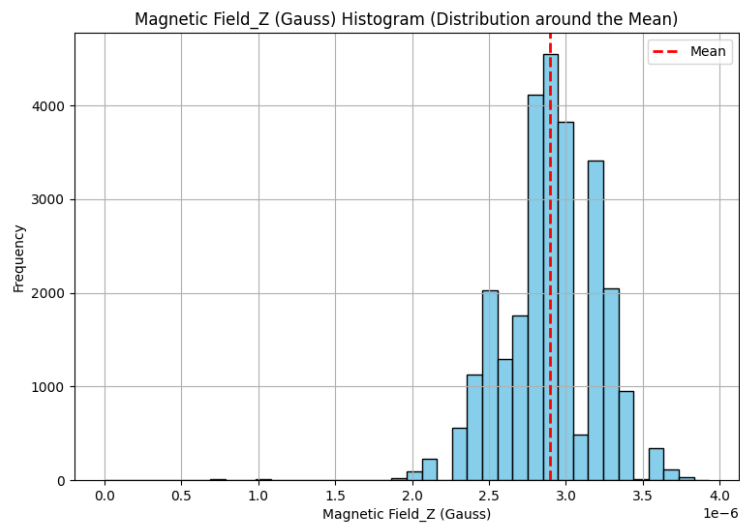
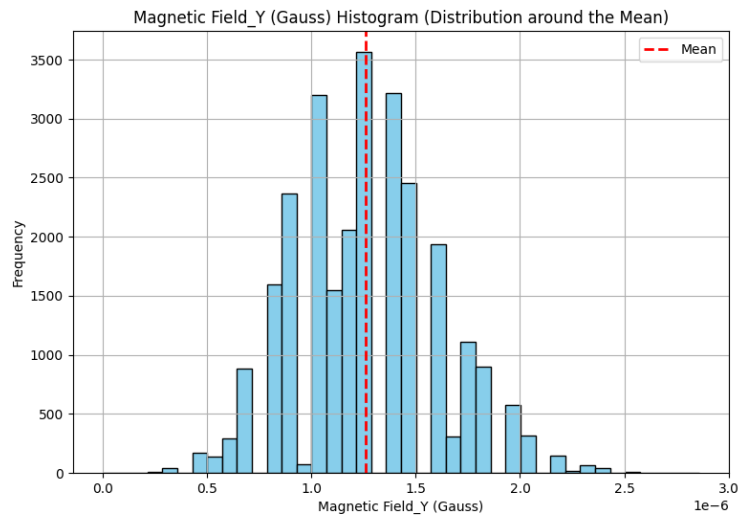
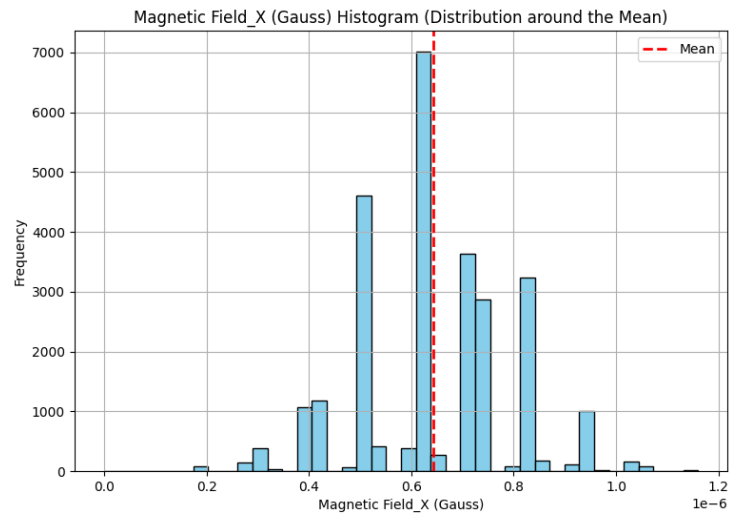


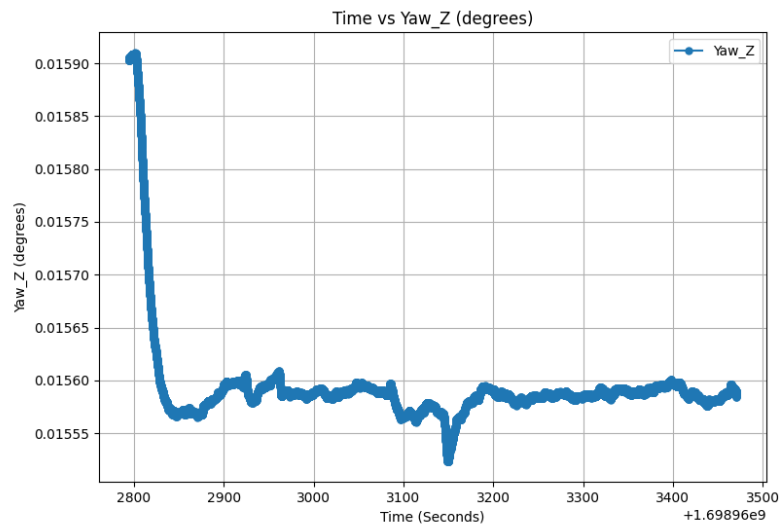
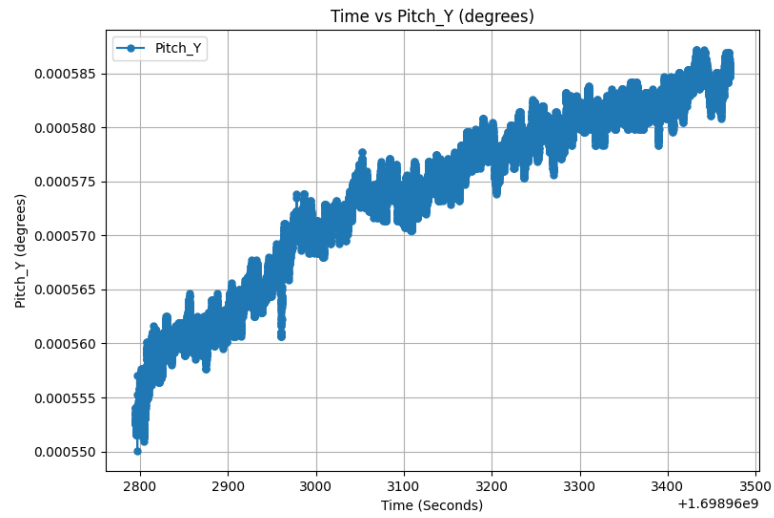
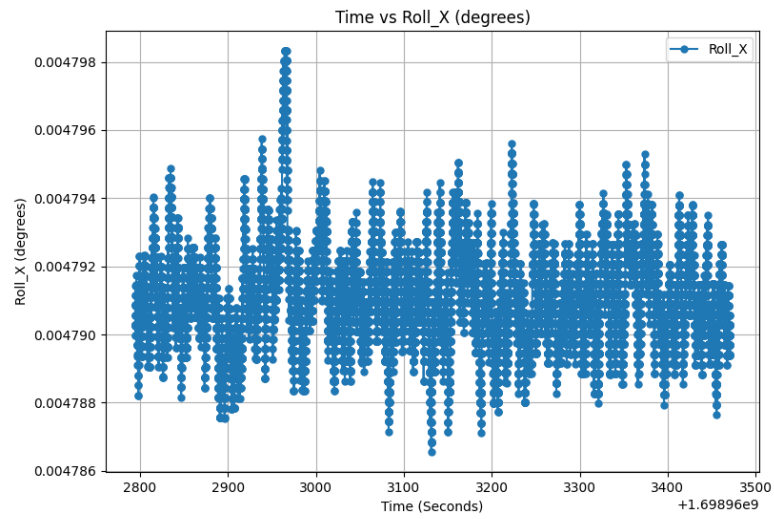


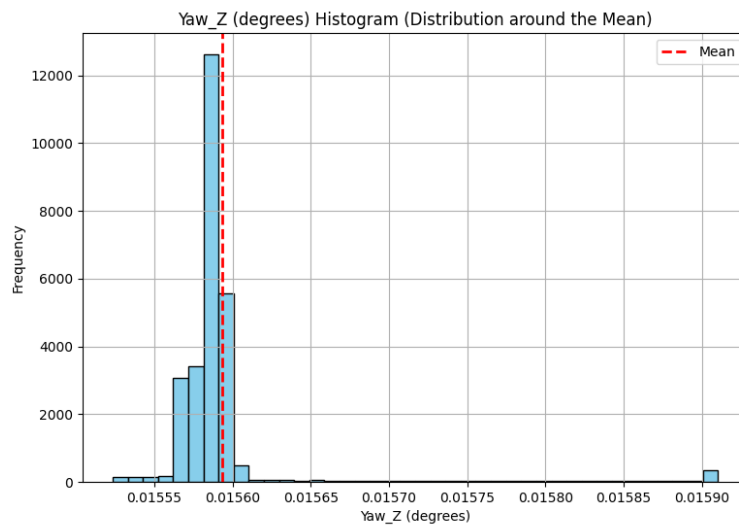
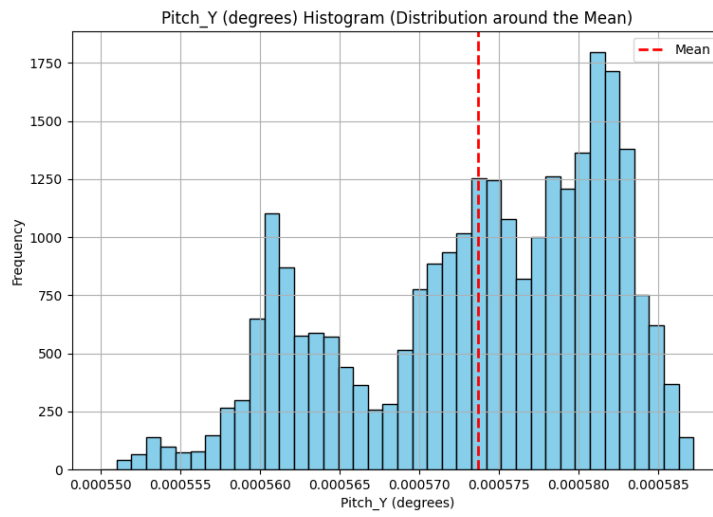
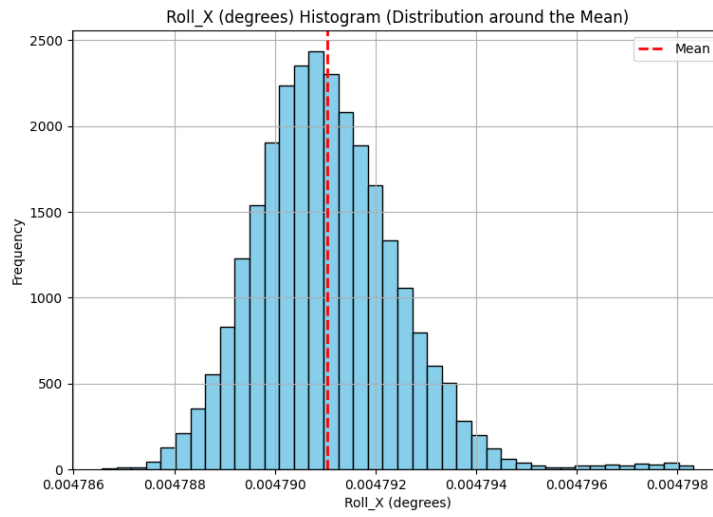












Stationary Data Statistics

<i>Gyroscope</i>	X	Y	Z
Mean	8.21702e-3	2.5459e-2	5.2158e-3
Median	8.219e-3	2.5457e-2	5.2155e-3
Standard Deviation	6.9716e-4	9.8363e-4	6.6629e-4
Distribution	Gaussian	Gaussian	Gaussian

<i>Accelerometer</i>	X	Y	Z
Mean	3.3917e-1	2.5122e-1	4.3523e-1
Median	3.39e-1	2.51e-1	4.35e-1
Standard Deviation	2.557e-2	1.5892e-2	2.1345e-2
Distribution	Gaussian	Gaussian	Lognormal

<i>Magnetometer</i>	X	Y	Z
Mean	6.4336e-7	1.2595e-6	2.8963e-6
Median	6.1999e-7	1.26e-6	2.91e-6
Standard Deviation	1.5018e-7	3.4506e-7	3.0975e-7
Distribution	Gamma	Gamma	Gaussian

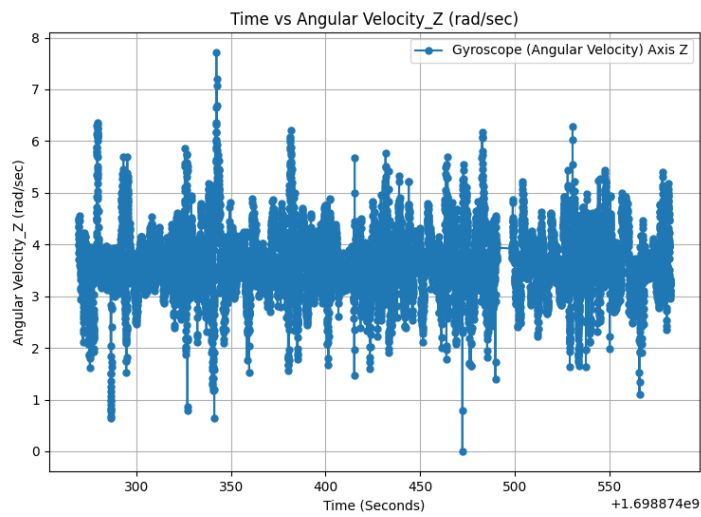
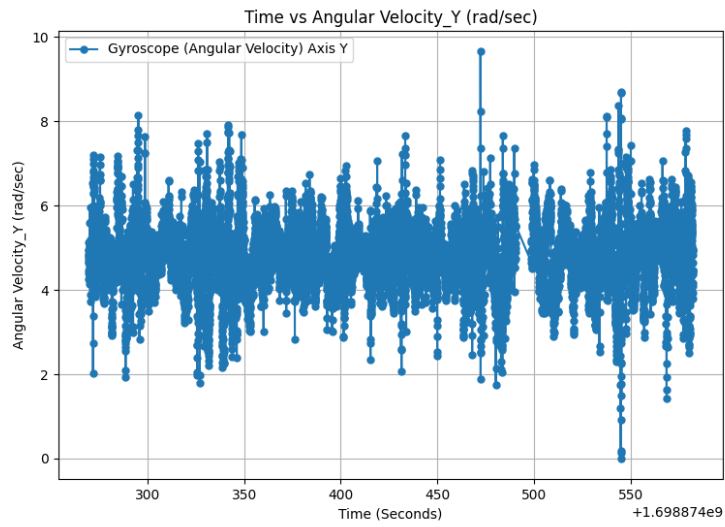
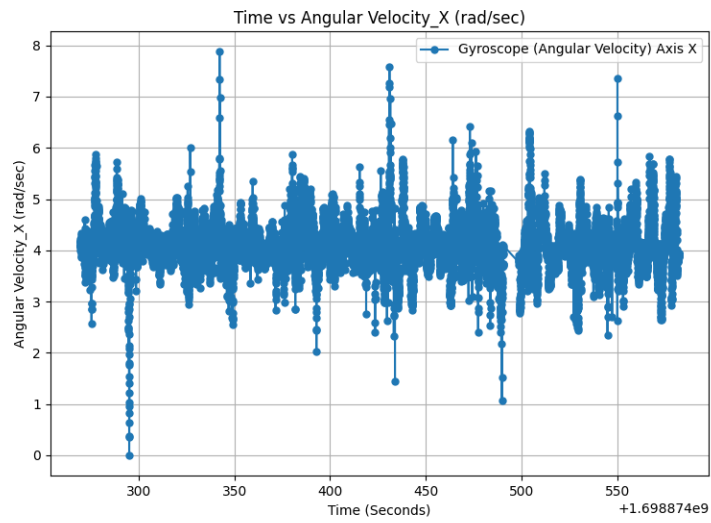
<i>Orientation</i>	Roll	Pitch	Yaw
Mean	4.791e-3	5.7369e-4	1.5593e-2
Median	4.79e-3	5.75e-4	1.5586e-2
Standard Deviation	1.413e-6	8.16e-6	4.9165e-5
Distribution	Lognormal	Weibull min	Gaussian

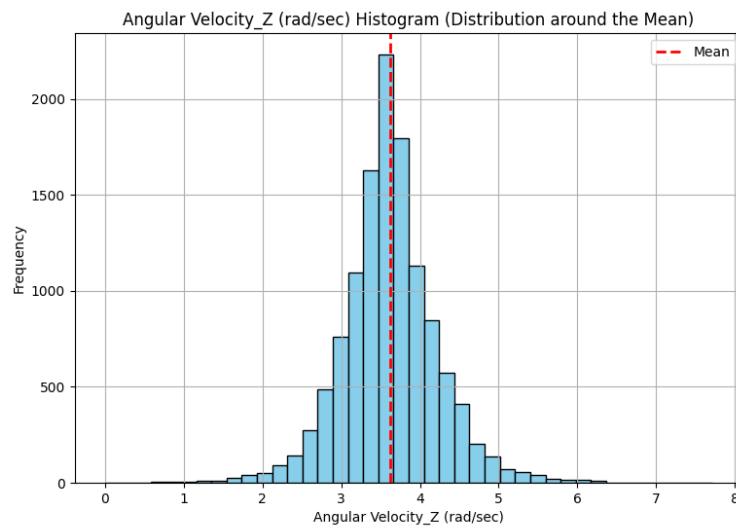
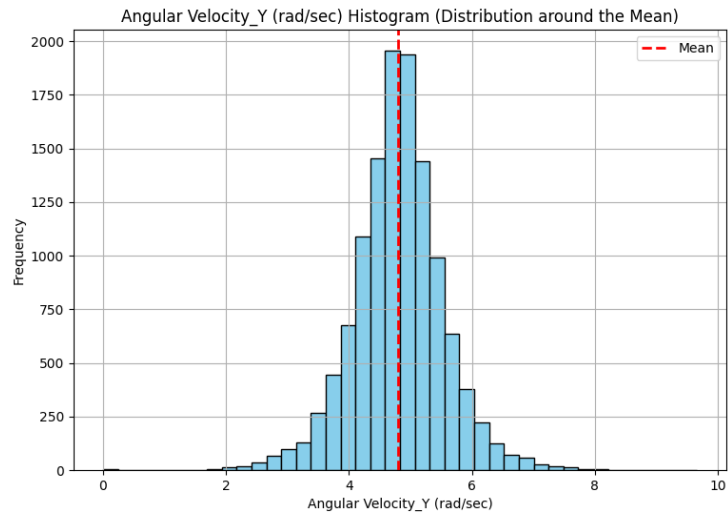
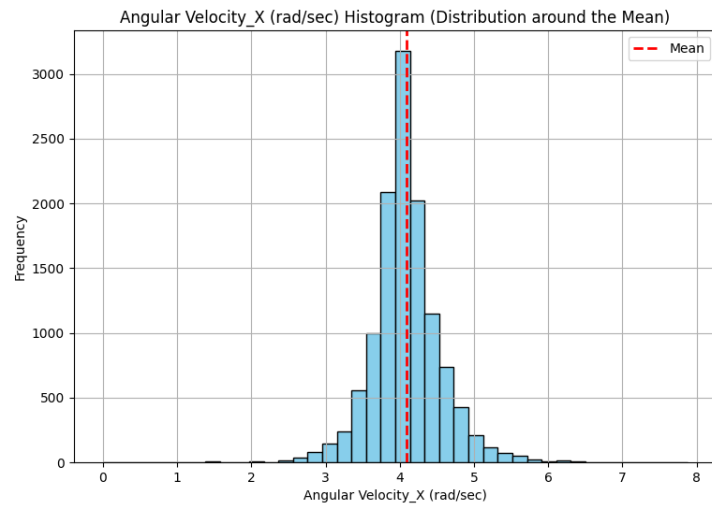
c. 5-minute Moving Data Analysis

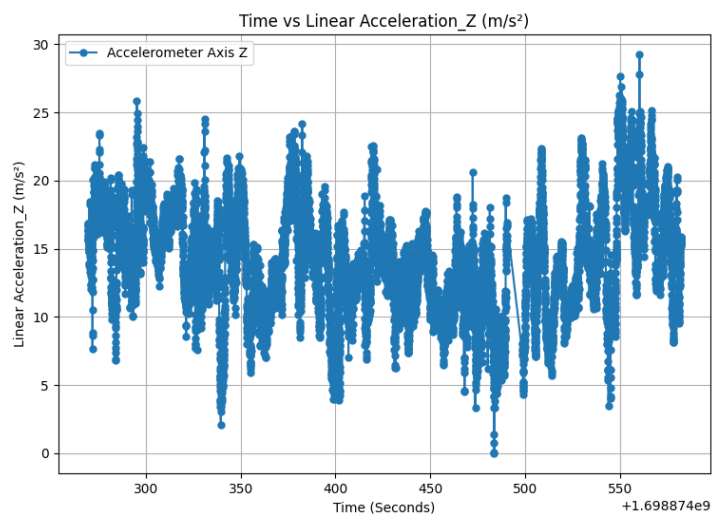
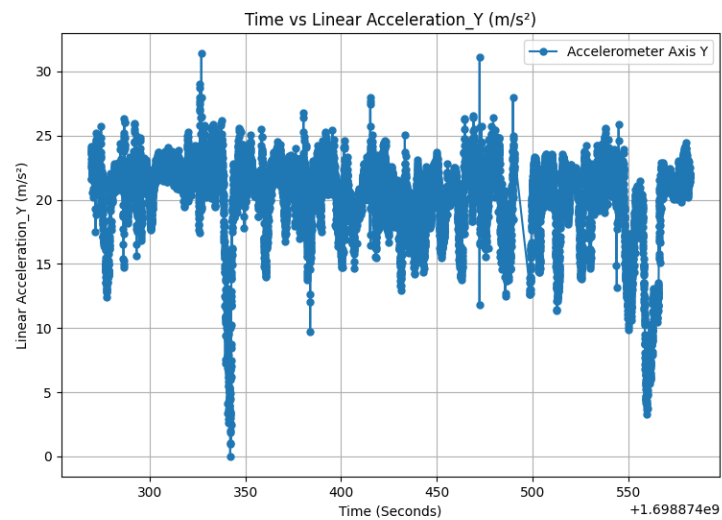
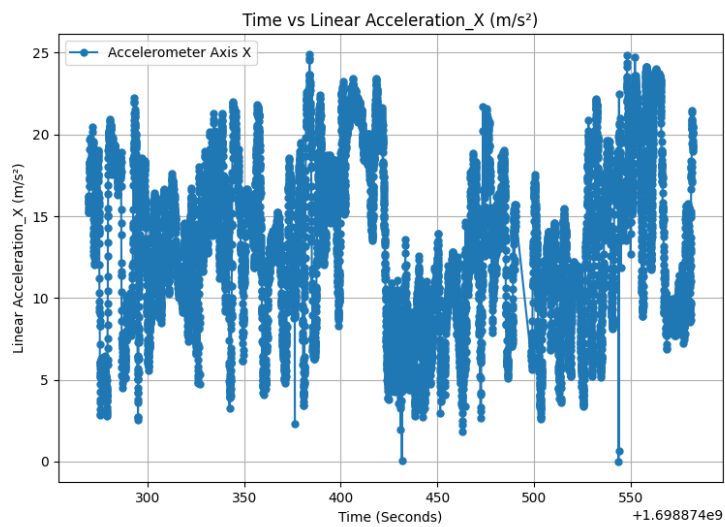
The dataset was collected on the first floor of West Village. The moving data set is critical for understanding the sensor's response to dynamic conditions, where the readings are influenced by a range of motions. The gyroscope statistics are compared to the stationary data, reflecting the variability in angular velocity during movement. The accelerometer statistics show increased mean and standard deviation, indicating the sensor's response to varying linear accelerations. The magnetometer statistics in the moving scenario have larger variations, likely due to changes in the surrounding magnetic field as the sensor moves. The orientation statistics reveal the IMU's ability to capture orientation changes, with the standard deviations reflecting the dynamic nature of the movement. The moving data set is most indicative of the sensor's performance in real-world, dynamic applications, highlighting its sensitivity to motion and orientation changes.

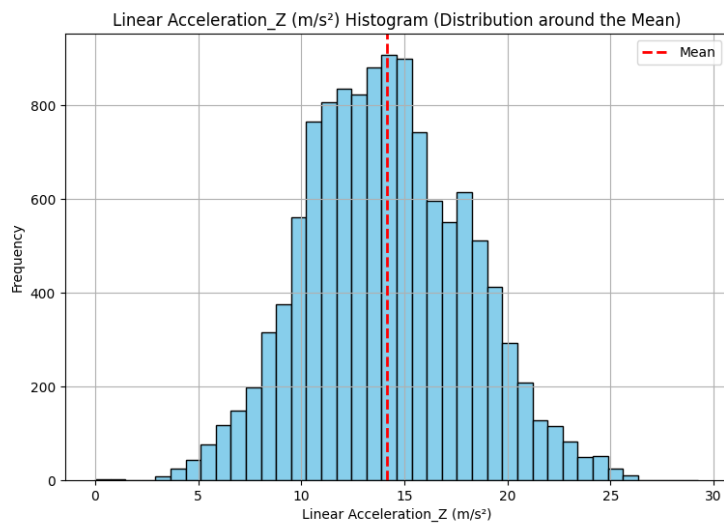
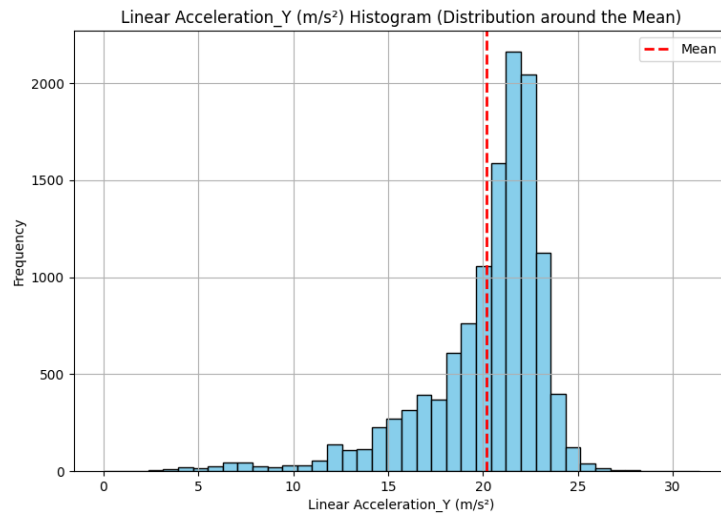
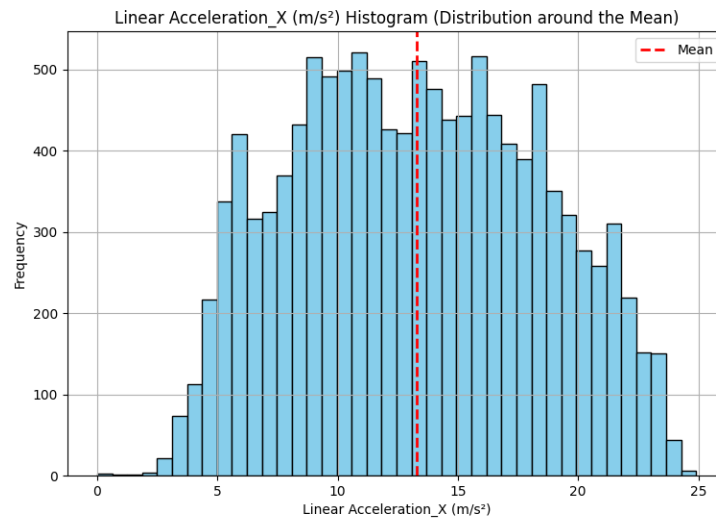
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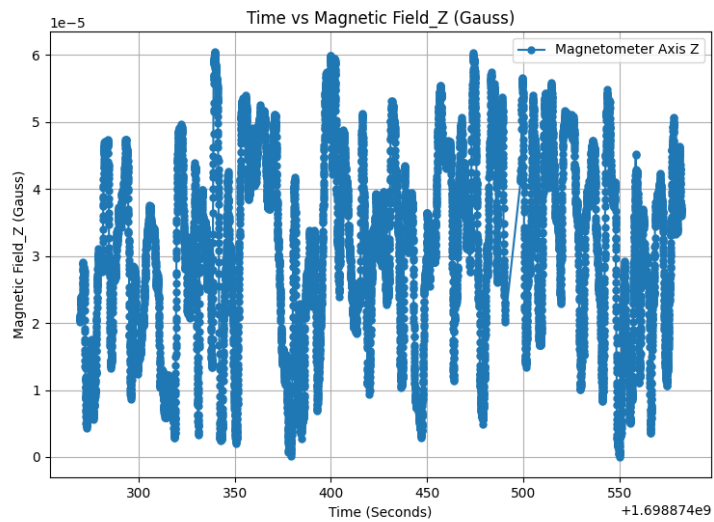
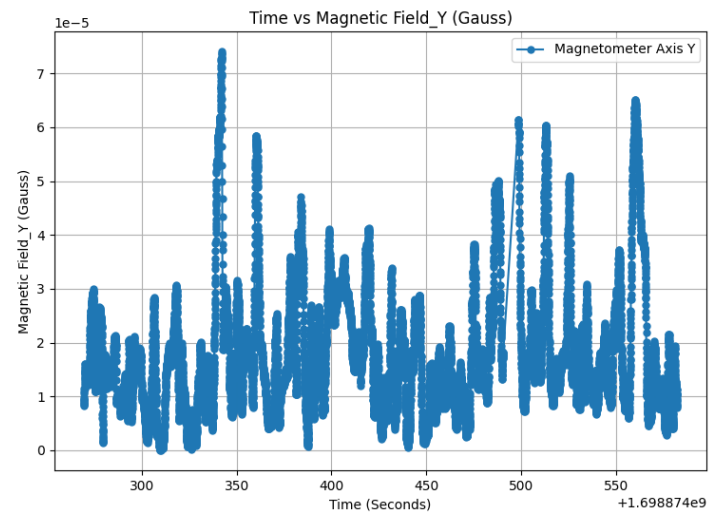
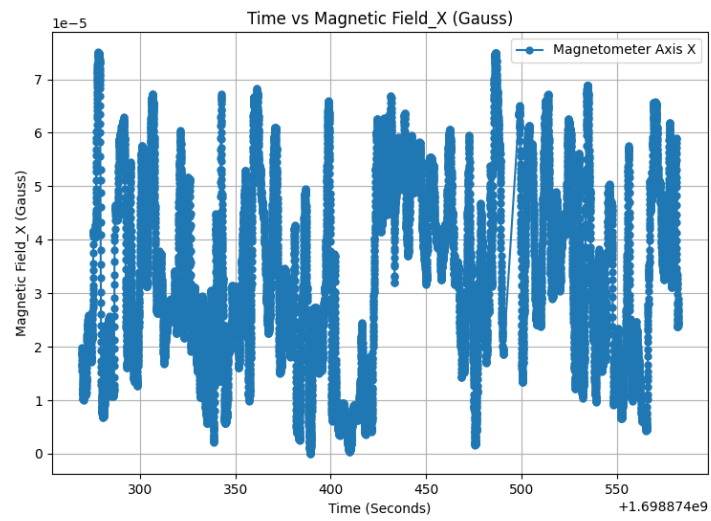
- **Dynamic Motion** - The most significant source of noise in moving data comes from the variability and unpredictability of the motion itself, which can introduce a wide range of accelerations and angular velocities.
- **Sampling Rate Issues** - If the sampling rate is not high enough to capture the dynamics of the movement accurately, this can lead to aliasing or missed nuances in the motion profile.
- **Magnetometer Interferences** - In moving scenarios, the magnetometer readings can be particularly susceptible to interference from nearby metallic objects or electromagnetic sources.

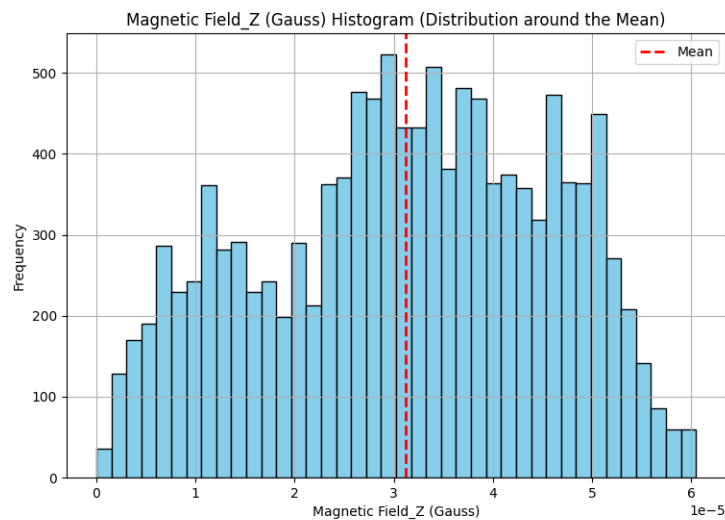
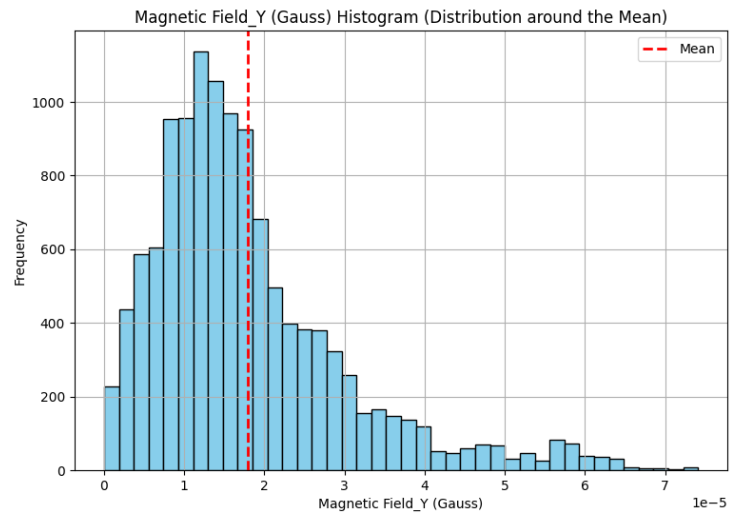
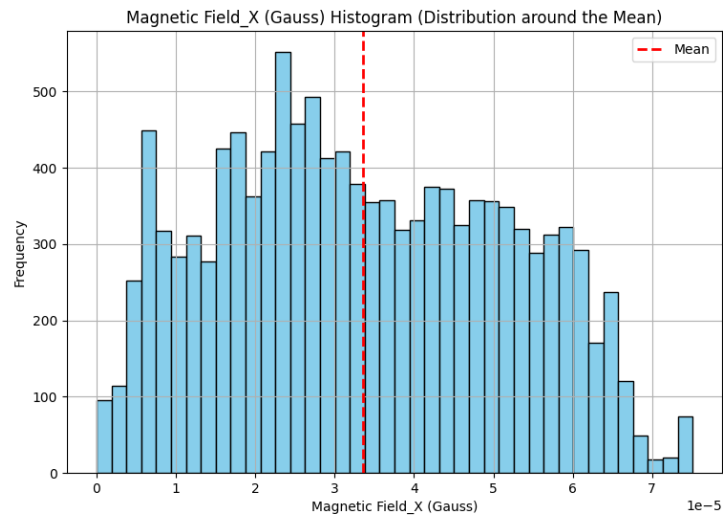


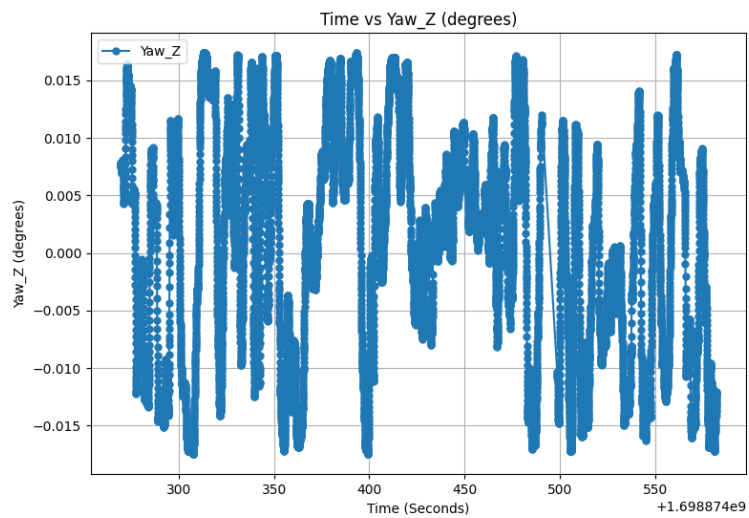
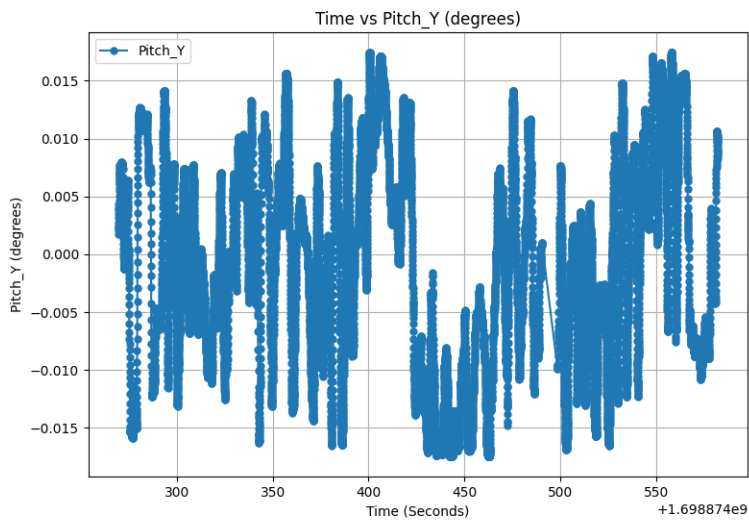
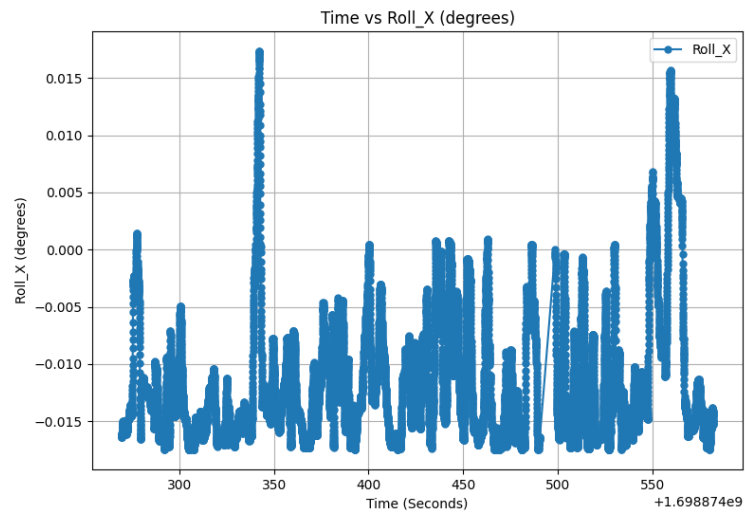


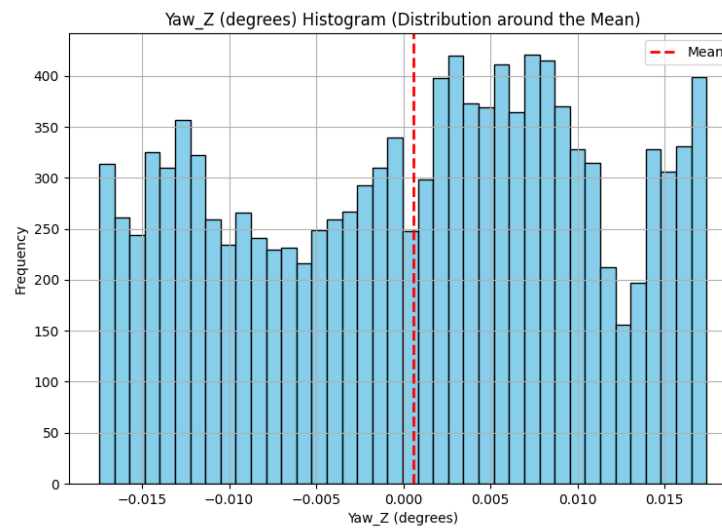
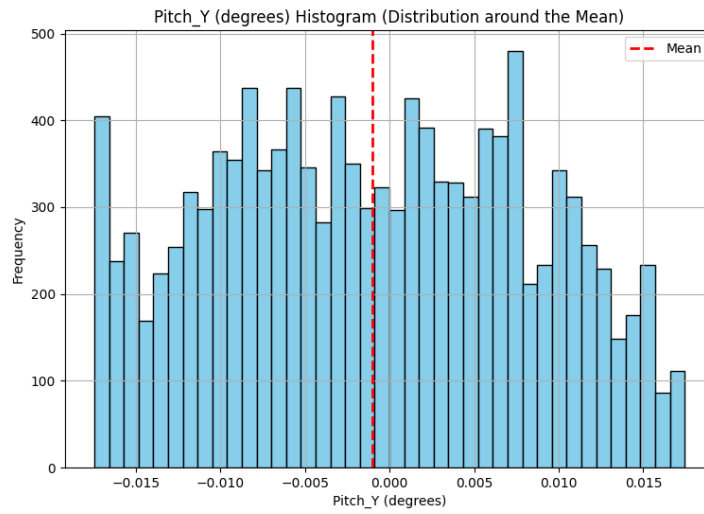
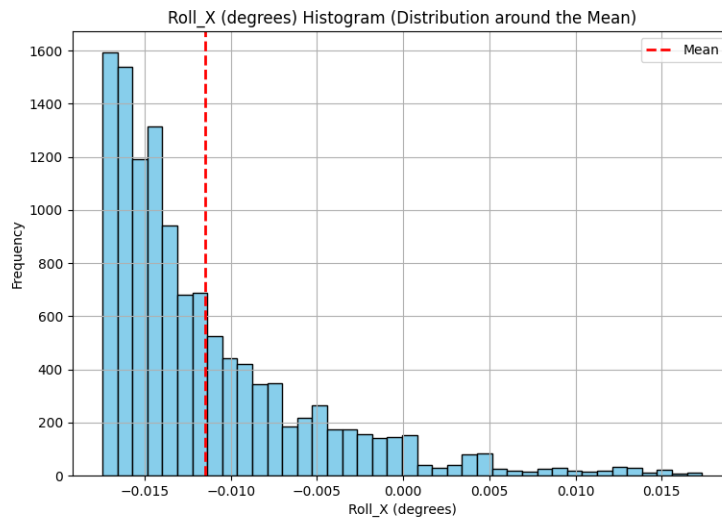












Motion:

1. The first motion can be seen in the plot of x axis where I was constantly moving the IMU in upward and downward motions.
2. The second motion can be seen at the beginning of the y axis time series plot where I did a swing of my arm from left to right.
3. The third motion can be seen in the plot of z axis where I did a twirl.

Moving Data Statistics

<i>Gyroscope</i>	X	Y	Z
Mean	4.08705	4.80546	3.61730
Median	4.05744	4.8114	3.601
Standard Deviation	0.47261	0.74174	0.6076
Distribution	Lognormal	Gaussian	Gamma

<i>Accelerometer</i>	X	Y	Z
Mean	13.28	20.1631	14.1347
Median	13.178	21.1305	13.9825
Standard Deviation	5.09046	3.3152	3.9024
Distribution	Weibull min	Gaussian	Gamma

<i>Magnetometer</i>	X	Y	Z
Mean	3.3549e-5	1.7919e-5	3.1179e-5
Median	3.195e-5	1.505e-5	3.204e-5
Standard Deviation	1.7849e-5	1.2234e-5	1.44e-5
Distribution	Weibull min	Lognormal	Gamma

<i>Orientation</i>	Roll	Pitch	Yaw
Mean	-0.01147	-0.001	5.579e-4
Median	-0.0136	-0.0011	1.749e-3
Standard Deviation	6.032e-3	9.1265e-3	9.9388e-3
Distribution	Weibull min	Gaussian	Gamma

Discussion

1. Noise in Short Stationary Data
 - The noise characteristics of the short stationary data were predominantly Gaussian, as indicated by the close alignment of median and mean values across the gyroscope and accelerometer axes. The gyroscope data exhibited mean values of 0.008217, 0.025460, and 0.005216, with corresponding medians of 0.008219, 0.025457, and 0.0052155, suggesting a symmetric distribution of noise. The accelerometer data also showed a similar pattern with mean values of 0.3392, 0.2512, and 0.4352 closely matching the median values. The standard deviations for these sensors were relatively low, indicative of a stable environment with minimal noise. However, the magnetometer data displayed a mix of gamma and normal distributions across different axes, pointing to varied noise influences. Overall, the data conformed largely to Gaussian noise, typical of white noise in electronic devices, with some minor environmental or electronic influences.

2. Alignment of Video and VectorNav Data

- To align the video with the VectorNav data, a synchronization process was employed. While one team member was capturing the video, another was recording the VectorNav data using a rosbag. This parallel recording ensured precise timing correlation between the two data sets. The key to this synchronization was the use of timestamps: the timestamp embedded in the rosbag message was matched with the timestamp from the video recording. By aligning these timestamps, we could accurately link the video footage to specific IMU data points. This method allowed us to effectively synchronize the two data streams, enabling us to capture and analyze critical moments in the video that corresponded with notable readings in the IMU data. This approach provided a comprehensive view of the IMU's performance and response in real-time conditions, as it offered a direct visual reference to the physical movements and orientations corresponding to the IMU data fluctuations.

3. Comparison with VN100 Datasheet

- Our measurements of the VN100 IMU's performance revealed that it underperformed compared to the specifications listed in the datasheet. The discrepancy can largely be attributed to the different testing environments. While the datasheet figures are obtained from tests in highly controlled, calibrated factory settings, designed to minimize external noise and vibrations, our data collection took place in a more dynamic, real-world environment. This environment was naturally susceptible to various external disturbances, such as ambient noise and vibrations, which are not present in factory test conditions. These external factors are critical in influencing the sensor's performance, leading to the observed variations. This comparison emphasizes the importance of considering environmental variables when evaluating sensor performance, as real-world conditions can significantly differ from idealized testing scenarios.

4. Possible Sources of Noise

- The subpar performance of our dataset compared to the VN100 datasheet could be attributed to various factors. Environmental vibrations, indicated by angle random walk and rate random walk noises from external sources like traffic or structural movements, along with Gaussian or white noise commonly associated with random fluctuations in electronic devices. Bias instability errors particularly in static inertial conditions, might have been influenced by temperature changes or electromagnetic interference. Additionally, human factors and setup inconsistencies such as improper calibration could have played a role. These elements combined suggest that a mix of environmental and operational conditions led to the observed discrepancies between our real-world measurements and the idealized datasheet specifications.

5. Allan Variance Plot Analysis

- Based on the Allan Variance plots, the noise characteristics and parameters observed in each plot, it seems plausible that the datasets from locations A, B, C, and D correspond to the 5th floor of ISEC, the ISEC basement, the 3rd floor of a wooden house, and the Snell library basement, respectively. It is likely that the wooden house and the ISEC 5th floor experienced greater environmental disturbances, possibly due to building movement or external traffic, leading to higher noise levels. While the ISEC and Snell library basements, being more insulated, exhibited lower noise levels and minimal external disturbances. In these environments, the predominant noise sources would include angle random noise and

bias instability, reflecting the influence of both the physical surroundings and the inherent sensor characteristics.

6. Optimal Conditions for Measuring Allan Variance

- The results offer valuable insights into the optimal conditions for measuring Allan variance of a sensor. Firstly, it's crucial to ensure that the sensor is well isolated from external disturbances, as these can significantly impact the readings and mask the true performance of the sensor. Conducting measurements over an extended period is also important, as this allows for a comprehensive understanding of long-term error propagation and sensor stability. Additionally, selecting the appropriate sampling rate is vital for accurate data analysis; it ensures that the data is representative of the sensor's performance under various conditions. Prior calibration of the sensor is another key step, as it helps correct any inherent noise characteristics, thus providing a more accurate assessment of the sensor's capabilities. In essence, creating a controlled, stationary environment for these measurements is imperative. This approach minimizes external noise influences and provides a clear baseline for the sensor's performance, enabling a more precise evaluation of its capabilities and limitations.

7. Characterizing a New Sensor

- To effectively characterize a new sensor with unknown performance, a two-way approach is advisable. Initially, data should be recorded in a controlled, quiet environment, ensuring minimal external disturbances. This setting will allow for the identification of the sensor's inherent noise and baseline performance characteristics. Subsequently, measurements should also be conducted in the sensor's intended operational conditions, which are likely to be more dynamic and challenging. This dual-environment testing strategy is crucial for distinguishing between the noise intrinsic to the sensor and that arising from external environmental factors. Analyzing data from both settings will provide a comprehensive understanding of the sensor's capabilities and limitations and offer insights into its expected performance in real-world applications.

Conclusion

The analysis of these three data sets demonstrates the VN 100's performance under different conditions. The 5-hour stationary data provides a baseline for sensor noise, the 5-minute stationary data captures short-term stability and minor environmental effects, and the 5-minute moving data illustrates the sensor's response to dynamic conditions. These insights are crucial for applications requiring precise motion tracking and orientation estimation in varying environments.